

The VIA eRev truck is a factory-new, electrified, extended range vehicle. This course is designed to familiarize operators with the truck's systems, as well as provide an overview of important truck driver safety information.

https://vimeo.com/136007732

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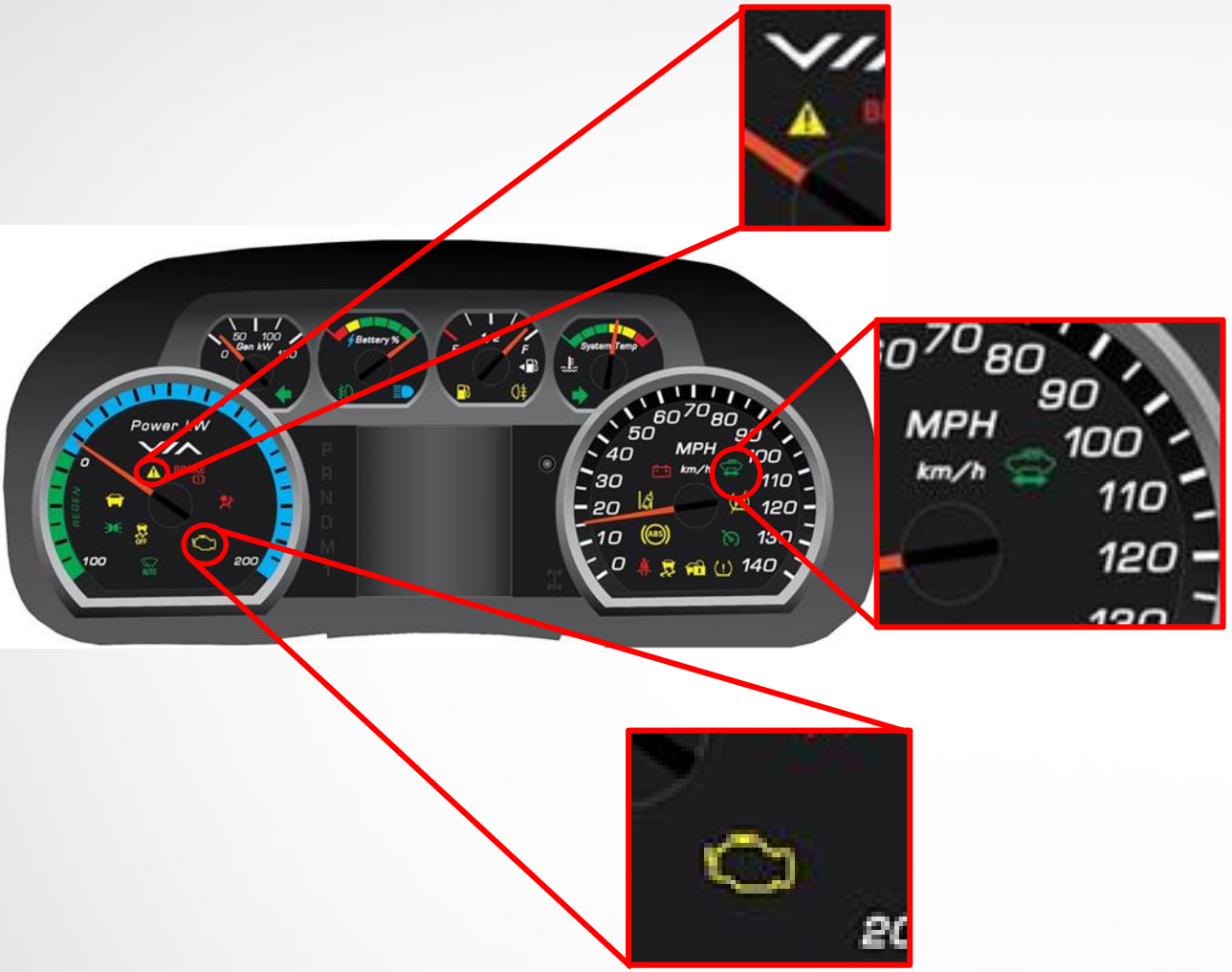
COURSE OVERVIEW

This course will focus on the following components:

- Driver interface
- Shift Selection
- Vapor Canister Purge “Purge Mode”
- Fluid fill locations
- Charging system
- Export power panel
- Low Voltage fuse center

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DRIVER INTERFACE



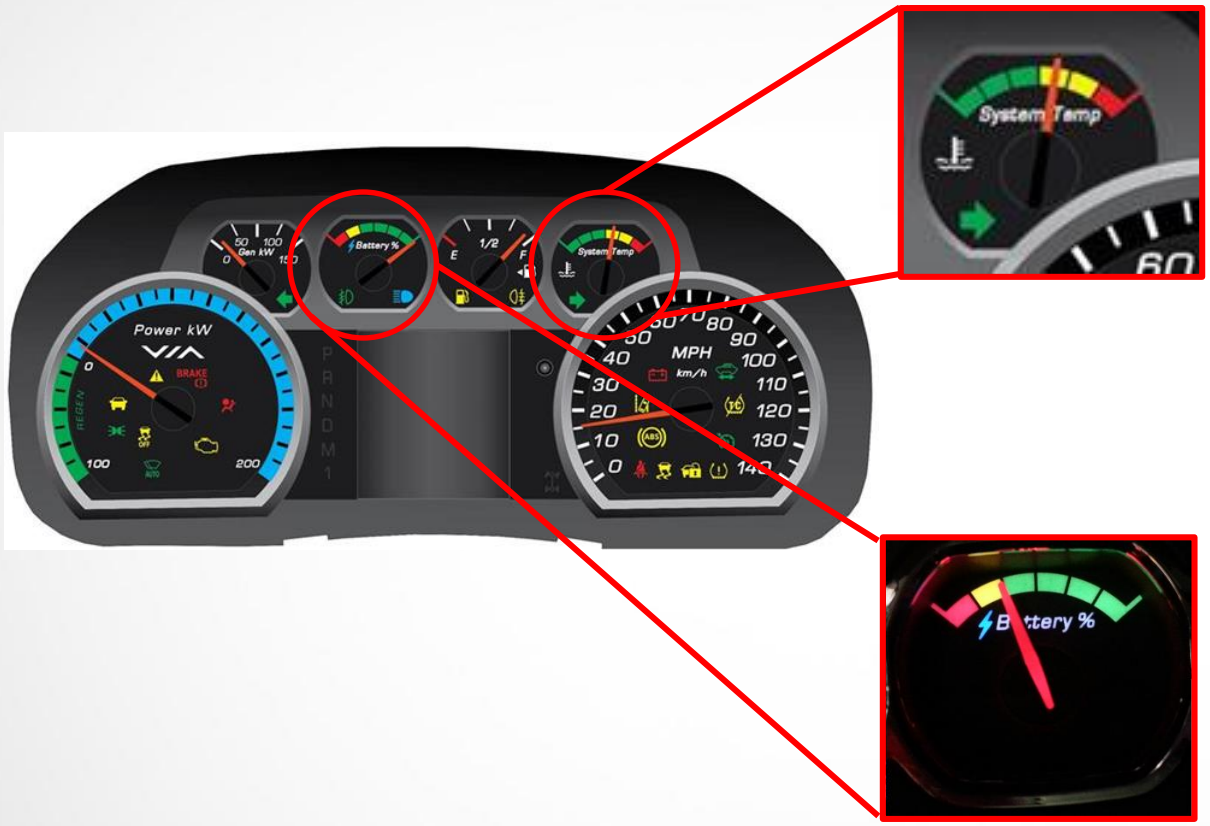
- The **Hybrid System Lamp** (HSL) illuminates solid when the VIA system detects a malfunction.
- The HSL will begin to flash on and off if it detects a severe malfunction.
- Depending on the severity, the vehicle performance may be automatically reduced to ensure safety of the driver and also the vehicle.

- The **Drive Ready indicator** will illuminate green when the vehicle is ready to drive.
- It is used while in battery electric startup mode to indicate all necessary systems are functioning and the vehicle is ready to drive.
- The indicator will stay on as long as the vehicle is active.

- The **Malfunction Indicator Lamp** (or MIL) illuminates amber when the vehicle is placed in ON/RUN, as a check to confirm functionality
- If the MIL stays on for more than 15 seconds on startup or while the engine is running, this indicates that the GM OBD 2 system has detected a malfunction.

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DRIVER INTERFACE

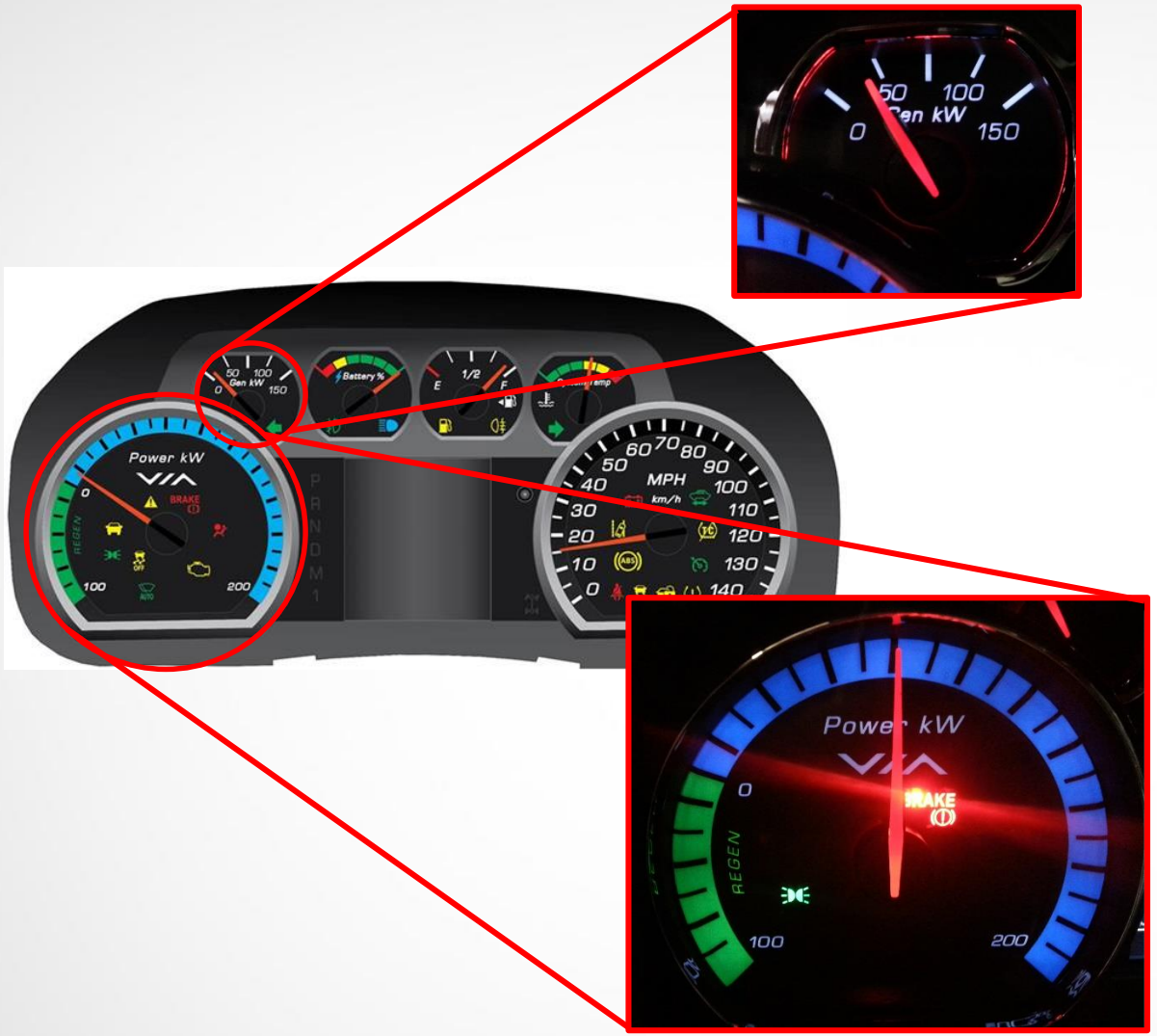


- The **System Temp Gauge** indicates the VIA cooling system temperatures.
- The system temp gauge is **Not** a constant variable gauge and only has three positons.
  - Normal – **Green**
  - Warm – **Yellow**
  - Hot – **Red**

- The **Battery Percent Gauge** indicates the charge in the High Voltage battery.
- When the EVSE is connected and charging is complete, the gauge will indicate 100% to reflect a fully-charged high voltage battery. May also show 80% charge if vehicle is in purge mode. See slide 7 for more information.
- As the vehicle is driven in battery electric mode, the percentage will reduce until the engine starts at 20%. At that point the vehicle power is sustained by the engine mounted generator.

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DRIVER INTERFACE



- The **Generator Kilowatt** gauge shows the output of the engine mounted generator.
- Shows high voltage output during export power mode.

- The **Power Kilowatt gauge** has a dual purpose.
- While accelerating or driving, the blue scale will show power usage.
- When coasting or braking, the green scale will show regenerative power recovery.

- **Note:** When battery state of charge is above 90% regenerative braking will be reduced to protect from over charging the battery.

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SHIFT SELECTION



The gear selector works the same as it would in any conventional vehicle (P is park, R is reverse etc.) because the vehicle does not have a transmission these actions are performed differently:

- Park is achieved with a module on the gearbox called a park pawl, which manually locks and unlocks the drivetrain.
- For Reverse, Neutral and Drive a signal is sent to the traction inverter to indicate direction of motor rotation.
- When the vehicle is placed into low gear (1 or M1) the APU (Auxiliary Power Unit) is enabled and will generate power according to demand and maintain current battery state of charge. This will also provide additional power during hill climbs and cold battery operation.

Note: When battery SOC is above 90% the engine will not fire until it drops below the 90%

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VAPOR CANISTER PURGE

The battery will not charge above 80% if a canister purge is required.

Due to C.A.R.B. regulations, if the vehicle has been used only in battery electric mode or the combustion engine has not started for 5 days, the engine will automatically start when the vehicle is powered up to purge the vapor canister. EVAP canister purge is completed during a drive if the engine runs at sufficient load for at least 15 to 30 minutes. This may be achieved over any number of drives within 48 hours. If 15 to 30 minutes of sufficient engine load is not achieved within the 48 hour period, the timer will reset and 15 to 30 minutes will again be required. Sufficient engine load generally occurs when driving at speeds greater than 50 mph (i.e. highway driving). Once the canister is purged the vehicle may return to battery electric mode.

Note: Due to other C.A.R.B. regulations the battery state of charge has to stay within ± 1% of where the APU is started, this causes the engine to rev up and down to maintain this tolerance whether or not the vehicle is driving.

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FLUID FILL LOCATIONS



Power Electronics Coolant Reservoir

Generator, Traction Motor Coolant/Lubrication Reservoir

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CHECKING FLUIDS – GENERATOR / TRACTION MOTOR

Note: The Generator and Traction Motor fluid should be checked weekly while the vehicle is off and when the fluid is cold. This system uses Dexron VI ATF.

1. Flip the handle up, pull out the dipstick, and wipe it with a clean rag or paper towel.
2. Push it back in all the way, wait three seconds, and pull it back out again
3. Check both sides of the dipstick, and read the lower level. The fluid level must be between the 2 indicators on the dipstick. Be sure to keep the dipstick pointed down to get an accurate reading.
4. If the fluid level is in the acceptable range, push the dipstick back in all the way, then flip the handle down to lock the dipstick in place. If fluid level is low top off though the dipstick tube and check for leaks.



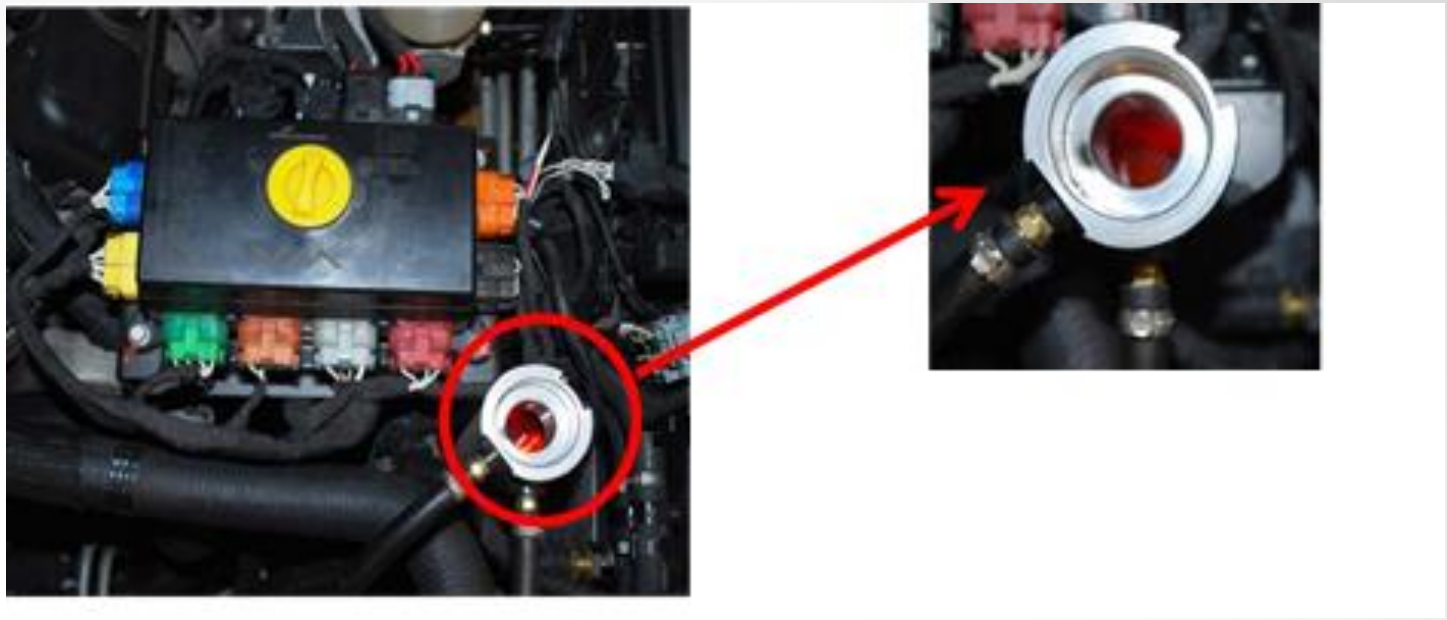


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POWER ELECTRONICS FLUID RESERVOIR

The power electronics fluid level should be checked weekly.

Warning: Steam and scalding liquids from a hot cooling system can blow out and cause serious burns. Never turn the reservoir pressure cap, when the cooling system is hot. Wait for the cooling system and reservoir pressure cap to cool.



1. Remove the reservoir pressure cap when the cooling system is no longer pressurized.

2. Push down the pressure cap as you turn it.

3. Remove the pressure cap.

4. Fill the reservoir with the proper mixture of 50% DEX-COOL and deionized water, 2 inches below the top of the reservoir. If fluid level is low inspect the vehicle for leaks.

5. Replace the pressure cap.

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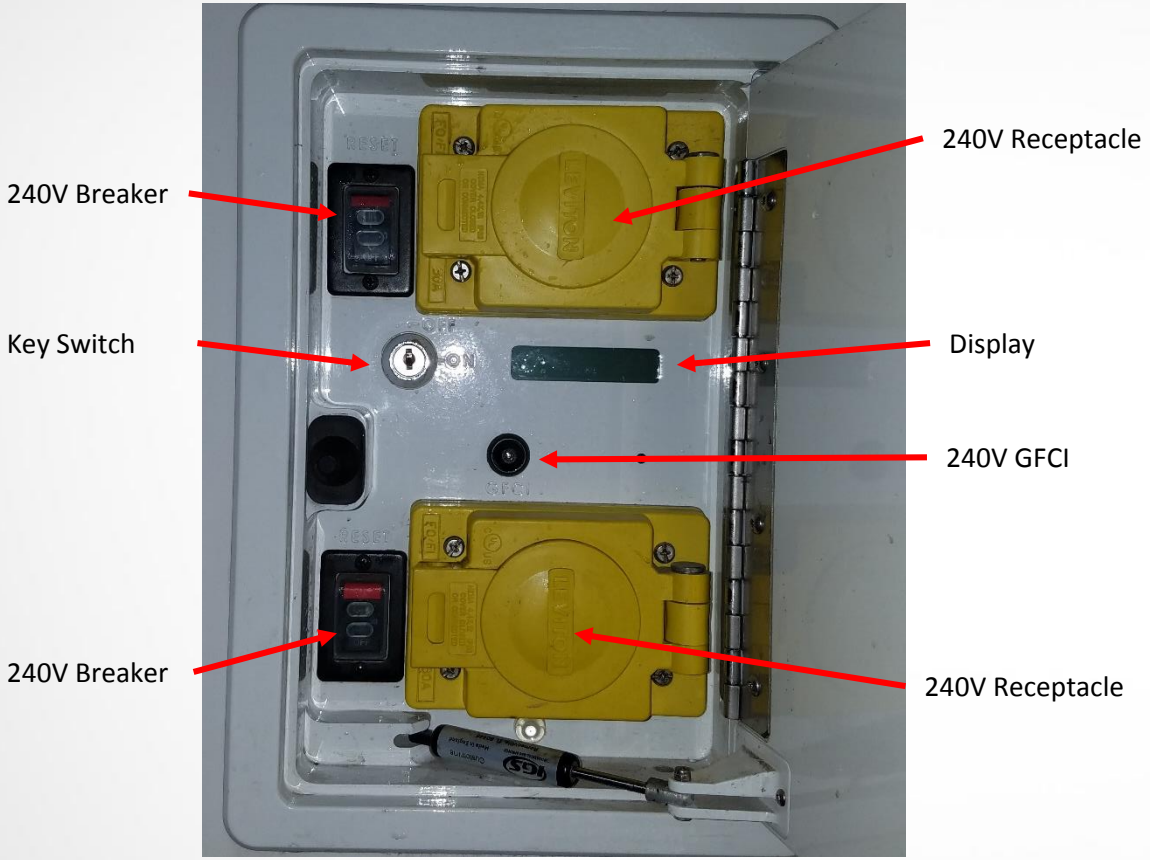
EVSE: ELECTRICAL VEHICLE SUPPLY EQUIPMENT PORT

- The Electric Vehicle Supply Equipment (EVSE) port is part of the Charging System and is located on the left front fender.
- Level 1 (120 volt) or level 2 (240 volt) EVSE charger can be used. This A/C voltage is sent to the Bi-directional charger. Not all chargers are the same so insure you are using one from the approved list.
- The power electronics pump is located near the center of the vehicle above the front brush guard. It will run as needed during the external charging process.
- Additional clicking sounds may be caused by the electrical devices used while charging and is part of the normal vehicle functions.
- The EVSE port indicator lamp will illuminate under these conditions:
  - Flashing green light indicates A/C power is flowing to charge the battery.
  - Solid green light indicates that the charge is complete.
  - Solid red light indicates a charging system fault and the vehicle is not currently charging.



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EXPORT PANEL

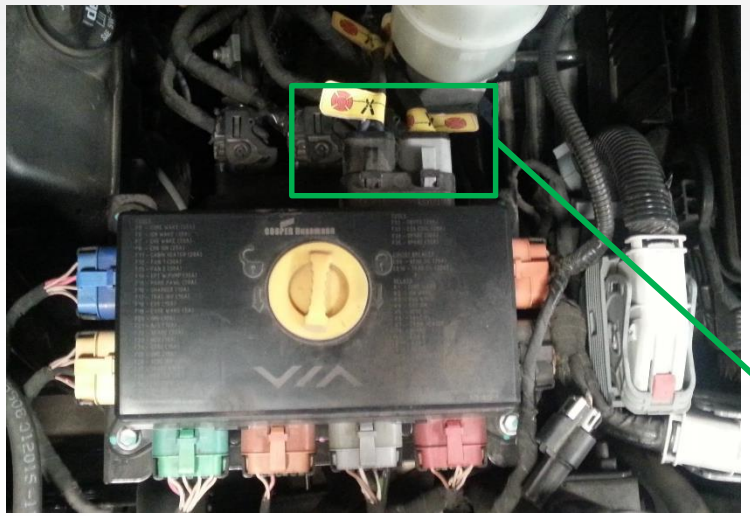


- The export power panel is located on the pass side of the bed between the cab and rear tire.
- It contains two 30 amp, 240 volt A/C outlets: One 3 prong and one 4 prong
- An adapter may be used to provide two 120v outlets.
- There is 14.4 kilowatts of total available power between both connectors.
- Export power may be used until the high voltage battery reaches a 20% state of charge. At that point an alarm will sound and the outlet will shut down.
- For continued export use, insert the vehicle key into the ignition and turn to the run position. The engine will start and drive the generator for continued power.

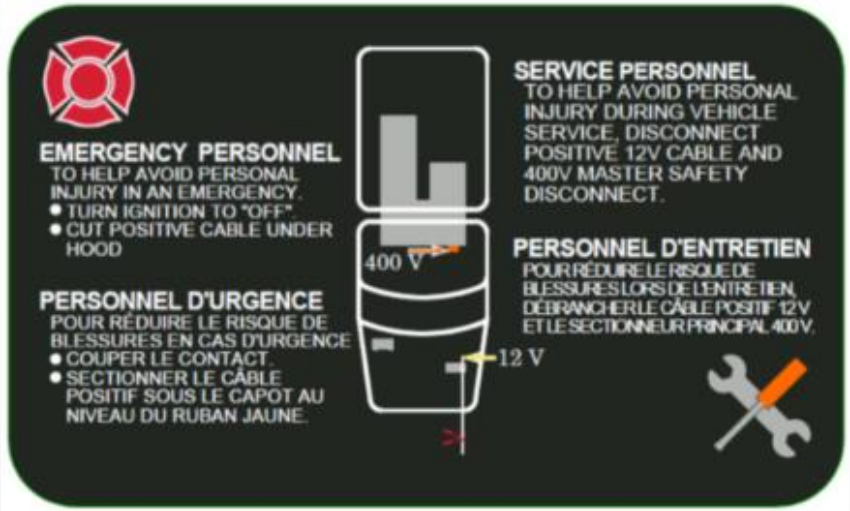
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LOW VOLTAGE FUSE CENTER/SERVICE DISCONNECT

Bussmann Fuse center



First Responder Info Sticker



- The low voltage fuse center is located under the hood on the drivers side.
- The fuse locations are inscribed on the fuse center cover.
- The black and Grey connectors are the main positive 12v feed to the VIA system and are one of the first responders manual disconnects.
- The high voltage manual service disconnect (MSD) is located on the front of the high voltage battery pack underneath the vehicle. This will be removed when servicing the high voltage system.

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